Software Requirements Specification

Facial Image Warping, Aging, and Expression Transformation System

CENG 384 DSP

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1. Introduction

This section provides the general context and purpose of the Facial Image Warping, Aging, and Expression Transformation system. It clearly explains the problem being addressed and the scope of the proposed software system.

1.1 Purpose of the System

The primary purpose of this software system is to provide a comprehensive Digital Signal Processing (DSP) tool for automated facial image manipulation and analysis. The system addresses the need for accessible, algorithm-driven facial modification by replacing complex, manual image editing processes. It solves this problem by offering an automated approach to facial landmark extraction, geometric warping for expression transformation (e.g., smile enhancement), and frequency-based age simulation, while simultaneously providing mathematical tools to evaluate these transformations quantitatively.

1.2 Scope of the System

The system operates within the application domain of digital image processing and computer vision. Its core functionalities include taking frontal face images (JPG, PNG) as input, detecting facial landmarks using pre-trained models, and applying geometric or frequency domain manipulations. The system allows users to alter facial expressions, simulate aging/de-aging, and analyze the results using frequency domain tools (such as 2D Fourier Transform and Magnitude Spectrum) and quantitative evaluation metrics (MSE, PSNR, SSIM).

2. System Overview

The Facial Image Warping, Aging, and Expression Transformation System is a desktop-based Digital Signal Processing application designed to perform automated facial image manipulation and quantitative analysis. The system accepts frontal face images as input and processes them through a multi-stage pipeline that includes face detection, landmark extraction, geometric and frequency-domain transformations, and result evaluation.

The system is composed of the following major modules:

- Image Input & Preprocessing Module: Handles image upload, validation, face detection, cropping, normalization, and format conversion.
- Facial Landmark Detection Module: Detects and visualizes key facial landmarks using pre-trained models such as Dlib or MediaPipe Face Mesh.
- Expression Manipulation Module: Applies geometric warping techniques to simulate facial expression changes such as smile enhancement, eyebrow adjustment, and face slimming.

- Aging & De-Aging Simulation Module: Uses frequency domain manipulation to simulate wrinkle formation (aging) and skin smoothing (de-aging).
- Frequency Domain Analysis Module: Computes and visualizes the 2D Fourier Transform and magnitude spectrum, and performs spectral energy analysis.
- Quantitative Evaluation Module: Calculates MSE, PSNR, and SSIM metrics to objectively evaluate transformation quality.
- AI-Supported Extensions Module: Provides optional AI-based expression transfer, age estimation, real-time webcam processing, and model comparison capabilities.
- Visualization & User Interface Module: Presents before/after comparisons, adjustable parameter controls, and interactive image inspection tools.
- The system is intended to operate on standard personal computers running Windows, macOS, or Linux operating systems, without requiring specialized hardware for its core functionality.

The primary objectives of the system are to automate facial image analysis and manipulation through DSP algorithms, to provide intuitive visualization tools for both spatial and frequency domain results, and to allow quantitative evaluation of all applied transformations.

3. Functional Requirements

This section describes the specific functions and behaviors that the system must provide. Requirements are organized by system module. Each requirement is written in a clear and testable format following the structure: The system shall + action.

3.1 Image Input & Preprocessing Module

FR-1.1: The system shall allow users to upload a frontal face image in supported formats such as JPG or PNG through the user interface for facial analysis processing.

FR-1.2: The system shall validate the uploaded image by checking its file format and resolution to ensure that it meets the minimum requirements necessary for facial analysis.

FR-1.3: The system shall display the uploaded image to the user before the processing stage begins in order to allow the user to verify that the correct image has been selected.

FR-1.4: The system shall automatically detect the face region within the uploaded image using a face detection algorithm.

FR-1.5: The system shall crop the detected face region from the original image and normalize the face area to prepare it for further image processing and analysis.

FR-1.6: The system shall convert the processed face image into grayscale format when required in order to support frequency-domain or texture analysis algorithms.

FR-1.7: The system shall resize processed face images to a predefined standard resolution to ensure consistent and reliable evaluation across different images.

3.2 Facial Landmark Detection Module

FR-2.1: The system shall detect facial landmarks such as eyes, nose, mouth, and facial contour points using a pre-trained facial landmark detection model such as Dlib, MediaPipe Face Mesh, or an AI-based deep learning model.

FR-2.2: The system shall visually display the detected facial landmarks by overlaying the landmark points on the processed face image.

FR-2.3: The system shall allow users to toggle the visualization of facial landmarks, enabling them to show or hide the landmark points on the displayed image.

FR-2.4: The system shall export the detected facial landmark coordinates so that they can be used in further geometric transformations or facial analysis processes.

3.3 Expression Manipulation Module (Geometric Warping)

FR-3.1: The system shall modify mouth landmarks to simulate a stronger smile.

FR-3.2: The system shall perform geometric warping based on facial landmark displacement.

FR-3.3: The system shall adjust eyebrow landmarks vertically while maintaining natural facial proportions.

FR-3.4: The system shall increase the horizontal lip distance while preserving texture continuity.

FR-3.5: The system shall modify cheek and jaw landmarks to create a slimmer facial shape.

FR-3.6: The system shall implement triangle-based warping and ensure smooth pixel interpolation while preventing visual artifacts.

3.4 Aging & De-Aging Simulation Module

FR-4.1: The system shall simulate facial aging using frequency domain manipulation.

FR-4.2: The system shall increase high-frequency components to simulate wrinkle formation.

FR-4.3: The system shall reduce skin smoothness using filtering techniques during aging simulation.

FR-4.4: The system shall simulate de-aging by reducing high-frequency components and smoothing skin regions.

FR-4.5: The system shall preserve important facial edges during the de-aging process.

FR-4.6: The system shall support optional AI-based aging models and allow comparison with frequency-based aging results.

3.5 Frequency Domain Analysis Module

FR-5.1: The system shall compute the 2D Fourier Transform of both the original image and the transformed image to enable frequency domain analysis.

FR-5.2: The system shall generate and display the magnitude spectrum of the images in the frequency domain using logarithmic scaling.

FR-5.3: The system shall analyze spectral energy distribution by computing the total spectral energy as well as high-frequency and low-frequency energy components.

FR-5.4: The system shall perform a comparative analysis between the original and transformed images in the frequency domain and present the results through visual comparisons and numerical tables.

3.6 Quantitative Evaluation Module

FR-6.1: The system shall compute the Mean Squared Error (MSE) between the original image and the transformed image to measure pixel-level differences.

FR-6.2: The system shall compute the Peak Signal-to-Noise Ratio (PSNR) based on the calculated MSE value to evaluate the quality of the transformed image.

FR-6.3: The system shall compute the Structural Similarity Index (SSIM) to evaluate the perceptual similarity between the original and transformed images.

FR-6.4: The system shall display all evaluation metrics in a structured results table and allow exporting the results as CSV or PDF files.

3.7 AI-Supported Functional Extensions Module

FR-7.1: The system shall use an AI-based model to transfer facial expressions from a reference image to the input face image.

FR-7.2: The system shall estimate the apparent age of the face image before and after transformation using an AI-based age estimation model.

FR-7.3: The system shall support real-time facial manipulation using webcam input.

FR-7.4: The system shall allow users to compare AI-based transformation results with traditional transformation techniques.

FR-7.5: The system shall allow users to select different AI models for facial transformation operations.

FR-7.6: The system shall display processing status information while AI-based transformations are being executed.

3.8 Visualization and User Interface Module

FR-8.1: The system shall display the original and transformed images side by side for visual comparison.

FR-8.2: The system shall allow users to adjust transformation parameters such as expression intensity, aging level, and smoothing level using interface controls.

FR-8.3: The system shall allow users to save processed images and analysis results.

FR-8.4: The system shall allow users to interactively zoom, pan, and inspect processed images.

FR-8.5: The system shall provide slider-based controls for adjusting transformation parameters.

FR-8.6: The system shall update the visualization automatically when transformation parameters are changed.

4. Non-Functional Requirements

This section defines the quality attributes and operational constraints of the Facial Image Warping, Aging, and Expression Transformation System. These requirements describe how well the system performs its functions, rather than what functions it performs.

4.1 Performance Requirements

The system shall process facial transformations and analyses within acceptable time limits to provide a responsive user experience.

- The system shall complete facial landmark detection within 2 seconds of image upload on standard hardware.
- The system shall apply geometric warping transformations (smile enhancement, eyebrow raising, lip widening, face slimming) and display the result within 3 seconds.
- The system shall compute the 2D Fourier Transform and display the magnitude spectrum within 3 seconds of a transformation being applied.
- The system shall calculate MSE, PSNR, and SSIM evaluation metrics and display the results table within 2 seconds.
- The system shall export results as CSV or PDF within 5 seconds of the user initiating the export action.
- The system shall load and display the uploaded image in the preview area within 1 second of a valid file being selected.

4.2 Usability Requirements

The system shall provide a clear, intuitive interface that allows users to perform facial image transformations and analyses without requiring prior expertise in image processing.

- The system shall present all transformation controls (sliders, toggle buttons) in a clearly labeled and logically grouped layout.
- The system shall display the original and transformed images side-by-side at all times, allowing users to directly compare results without additional navigation.
- The system shall allow users to adjust transformation intensity via interactive sliders, with changes reflected on the preview image in real time or within 1 second.
- The system shall provide a landmark visualization toggle (show/hide) that activates instantly without reprocessing the image.
- The system shall display all quantitative evaluation metrics (MSE, PSNR, SSIM) in a structured, readable results table that is visible without scrolling on a standard 1080p display.
- The system shall provide descriptive error messages in plain language when an invalid file format or unsupported image resolution is uploaded.
- The system shall allow a first-time user to complete a full transformation and view evaluation results within 5 minutes without external assistance.

4.3 Reliability Requirements

The system shall operate stably and produce consistent, reproducible results across repeated uses under normal operating conditions.

- The system shall produce identical transformation and evaluation results when the same image and the same parameter settings are applied across separate sessions.
- The system shall validate uploaded images before processing and reject unsupported file formats (non-JPG, non-PNG) with an appropriate error message, without crashing.
- The system shall handle cases where face detection fails gracefully, notifying the user with a clear message rather than proceeding with incorrect processing.
- The system shall preserve the original uploaded image in memory throughout the session, ensuring that the before-after comparison always reflects the unmodified source.
- The system shall not lose unsaved results or crash due to repeated transformation operations within a single session.

- The system shall recover to a stable state if an optional AI-based module (GAN or diffusion model) fails to load, falling back to the frequency-based processing pipeline.

5. System Constraints

This section defines the practical engineering limitations that affect the development and operation of the Facial Image Warping, Aging, and Expression Transformation System. These constraints are related to budget, hardware resources, computing capacity, and technology choices.

5.1 Budget Constraints

The system shall be developed with minimal cost by leveraging open-source tools and freely available resources throughout the entire development lifecycle.

- The system shall be developed primarily using open-source tools, libraries, and frameworks in order to minimize development costs.
- The project shall avoid the use of commercial or paid software licenses whenever possible.
- Pre-trained models used for facial landmark detection (e.g., Dlib, MediaPipe) shall be sourced from freely available open-source repositories.
- Optional AI-based modules (GAN or diffusion models) shall only utilize freely available pre-trained weights and open-source implementations.

5.2 Hardware Constraints

The system shall be designed to run on standard personal computers commonly available in academic environments, without requiring specialized or high-end hardware.

- The system shall operate on standard personal computers without requiring specialized hardware such as dedicated GPUs for core functionality.
- Minimum RAM requirement: 8 GB.
- Minimum processor requirement: Dual-core CPU at 2.0 GHz or above.
- Minimum storage requirement: 2 GB of available disk space for application files, temporary image data, and exported results.
- A display with a minimum resolution of 1280x720 pixels shall be supported for proper rendering of the side-by-side comparison interface.
- Optional AI-based processing modules may benefit from a dedicated GPU, but shall not require one for the system to function.